FHENumpy Test Report

- Mathematical Functions
  - Triangular Function
  - Hyperbolic function
  - Arithmetic operations
  - Rounding
  - Addition, Multiplication, and subtraction functions on an axis
  - Exponentials and Logarithms
  - Comparison Function
  - Miscellaneous
- Linear Algebra
  - Matrix and vector products
  - Norms and other numbers
  - Solving equations and inverting matrices
  - Miscellaneous
- Statistics
- Utility Functions
- Array Operations

> Hardware Configuration：CPU 13th Gen Intel(R) Core(TM) i5-13400 2.50 GHz RAM 16GB

MatFmatics Function

Trigonometric functions

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to perform 10,000 operations)		
					Scalar	Vector (length 10)	Vector (length 100)
1	sin	Sine function	15	FHENumpy	92.033 ms	122.735 ms	258.555 ms
				Numpy	5.426 ms	3.357 ms	7.006 ms
2	cos	Cosine function	15	FHENumpy	91.840 ms	121.294 ms	265.006 ms
				Numpy	5.897 ms	3.411 ms	7.129 ms
3	tan	Tangent function	14	FHENumpy	95.931 ms	136.713 ms	325.767 ms
				Numpy	5.713 ms	3.615 ms	8.259 ms
4	arcsin	Inverse sine function	15	FHENumpy	108.087 ms	220.975 ms	632.823 ms
				Numpy	4.905 ms	3.296 ms	7.860 ms
5	arccos	Inverse cosine function	15	FHENumpy	107.499 ms	221.672 ms	658.347 ms
				Numpy	4.940 ms	3.529 ms	6.818 ms
6	arctan	Inverse tangent function	16	FHENumpy	97.697 ms	230.074 ms	564.735 ms
				Numpy	4.230 ms	3.023 ms	6.906 ms
7	rad2deg	Convert angles from radians to degrees	16	FHENumpy	81.624 ms	98.552 ms	182.592 ms
				Numpy	4.348 ms	2.938 ms	3.647 ms
8	deg2rad	Convert angles from degrees to radians	16	FHENumpy	81.523 ms	98.617 ms	182.800 ms
				Numpy	4.447 ms	2.596 ms	3.644 ms
9	degrees	Convert angles from radians to degrees	16	FHENumpy	81.196 ms	98.266 ms	182.163 ms
				Numpy	4.597 ms	2.588 ms	4.002 ms
10	radians	Convert angles from degrees to radians	16	FHENumpy	81.555 ms	98.601 ms	181.658 ms
				Numpy	4.327 ms	3.158 ms	3.822 ms

11	unwrap	Expand by changing the difference between values to 2π complement (shift phase angle)	15	FHENumpy	--	99.784 ms	191.599 ms
				Numpy	--	79.035 ms	88.140 ms

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to perform 10,000 operations) (vector length 100)		
					Scalar, scalar	Vector, Vector	Scalar, vector
12	arctan2	Inverse tangent of x1/ x2, with the correct choice of quadrant	13	FHENumpy	112.423ms	821.136 ms	908.234 ms
				Numpy	7.554 ms	11.364 ms	14.512 ms
13	hypot	Pass in the "straight side" of a right triangle and return the hypotenuse	15	FHENumpy	103.623 ms	299.379 ms	373.357 ms
				Numpy	8.266 ms	7.001 ms	10.177 ms

Hyperbolic function

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to perform 10,000 operations)		
					Scalar	Vector (length 10)	Vector (length 100)
1	sinh	Hyperbolic sine function	13	FHENumpy	153.204 ms	317.828 ms	1052.448 ms
				Numpy	6.265 ms	3.515 ms	9.636 ms
2	cosh	Hyperbolic cosine function	13	FHENumpy	143.604 ms	319.103 ms	1055.051 ms
				Numpy	6.445 ms	3.688 ms	8.761 ms
3	tanh	Hyperbolic tangent function	16	FHENumpy	144.056 ms	340.154 ms	1062.669 ms
				Numpy	5.856 ms	3.582 ms	6.984 ms
4	arcsinh	Inverse hyperbolic sine function	14	FHENumpy	112.632 ms	136.418 ms	235.221 ms
				Numpy	5.540 ms	4.283 ms	17.297 ms
5	arccosh		13	FHENumpy	112.011 ms	136.050 ms	236.129 ms

		Inverse hyperbolic cosine function		Numpy	5.397 ms	3.521 ms	15.572 ms
6	arctanh	Inverse hyperbolic tangent function	15	FHENumpy	102.770 ms	125.870 ms	224.638 ms
				Numpy	4.115 ms	3.005 ms	13.311 ms

Arithmetic operations

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to perform 10,000 operations) (vector length 100)				
					Scalar, scalar	Vector, Vector	Scalar, vector	Scalar, plain text	Vector, plain text
1	add (+)	Returns the sum of the two input elements.	15	FHENumpy	97.870 ms	287.367 ms	357.862 ms	84.977 ms	192.374 ms
				Numpy	5.691 ms	3.045 ms	5.448 ms	6.020 ms	5.518 ms
2	sub (-)	Returns the difference between two input elements	15	FHENumpy	99.516 ms	281.686 ms	353.867 ms	85.663 ms	190.779 ms
				Numpy	5.669 ms	3.014 ms	5.425 ms	5.631 ms	5.632 ms
3	mul (*)	Returns the product of two input elements.	16	FHENumpy	99.590 ms	284.869 ms	358.761 ms	85.532 ms	185.233 ms
				Numpy	5.545 ms	3.157 ms	5.326 ms	5.616 ms	5.549 ms
4	div (/)	Divide two input elements and return x1/x2	15	FHENumpy	97.990 ms	286.985 ms	359.596 ms	85.539 ms	184.872 ms
				Numpy	7.996 ms	3.434 ms	6.322 ms	7.341 ms	5.600 ms
5	invers	Divide two input elements and return x2/x1	16	FHENumpy	96.088 ms	281.003 ms	354.968 ms	--	--
				Numpy	--	--	--	--	--

6	floor_divide	Return the maximum integer less than or equal to the input division	15	FHENumpy	129.806 ms	1030.216 ms	1135.647 ms	117.029 ms	934.004 ms
				Numpy	6.157 ms	11.987 ms	16.078 ms	6.358 ms	15.645 ms
7	fmod	Returns the remainder of a division	14	FHENumpy	130.020 ms	1020.674 ms	1111.096 ms	116.487 ms	918.384 ms
				Numpy	6.103 ms	9.003 ms	13.009 ms	6.187 ms	12.322 ms
8	mod (%)	Return the remainder of the division element	14	FHENumpy	131.182 ms	1034.521 ms	1134.061 ms	116.908 ms	938.874 ms
				Numpy	5.933 ms	9.600 ms	13.574 ms	6.064 ms	13.383 ms
9	Remainder	Return the remainder of the division element	14	FHENumpy	134.856 ms	1043.731 ms	1149.993 ms	118.088 ms	944.643 ms
				Numpy	5.994 ms	9.010 ms	13.559 ms	6.008 ms	12.142 ms
10	divmod	Return both quotient and remainder element-wise	14	FHENumpy	173.192 ms	1762.927 ms	1872.036 ms	158.560 ms	1685.353 ms
				Numpy	9.885 ms	17.006 ms	20.669 ms	9.715 ms	19.131 ms

No.	**Function Name**	**Function Description**	**Number of significant digits**	**Contrast to Numpy**	**Speed (average time to perform 10,000 operations)**		
					Scalar	**Vector (length 10)**	**Vector (length 100)**
11	pow (**)	Exponentiation (fractional powers)	15	FHENumpy	86.237ms	102.567 ms	187.868 ms
				Numpy	5.946 ms	8.778 ms	21.669 ms
12	float_power (**)	Floating point exponentiation	14	FHENumpy	108.069 ms	124.710 ms	225.414 ms
				Numpy	6.785 ms	5.347 ms	19.063 ms
13	reciprocal	Returns the reciprocal of the input element	16	FHENumpy	81.185 ms	99.029 ms	185.338 ms
				Numpy	4.891 ms	4.803 ms	4.882 ms
14	positive	Perform positive calculation on input elements	16	FHENumpy	83.879 ms	98.287 ms	183.906 ms
				Numpy	3.844 ms	2.384 ms	2.515 ms
15	negative (-)	Negate the input element	16	FHENumpy	81.588 ms	100.594 ms	184.960 ms
				Numpy	3.666 ms	2.580 ms	2.679 ms
16	decimal		15	FHENumpy	116.262 ms	449.432 ms	935.571 ms

		Returns the fractional part of the input element		Numpy	--	--	--
17	modf	Returns the fractional and integer parts of the input element	14	FHENumpy	156.572 ms	767.637 ms	1663.963 ms
				Numpy	8.978 ms	6.407 ms	8.903 ms

Rounding

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to perform 10,000 operations)		
					Scalar	Vector (length 10)	Vector (length 100)
1	rounding	Round the input element to the given number of decimal places (round down)	16	FHENumpy	119.125 ms	439.097 ms	896.396 ms
				Numpy	--	--	--
2	round	Round an array to a given number of decimal places (round to the nearest integer)	16	FHENumpy	245.140 ms	2709.753 ms	5839.001 ms
				Numpy	19.522 ms	10.717 ms	33.276 ms
3	rint	Round the input element to the nearest integer to (round to the nearest even number)	16	FHENumpy	233.130 ms	2699.861 ms	5828.588 ms
				Numpy	4.864 ms	5.786 ms	28.487 ms
4	fix	Round to zero to the nearest integer	14	FHENumpy	114.214 ms	442.143 ms	930.845 ms
				Numpy	31.438 ms	25.642 ms	30.916 ms
5	floor	Round to infinity towards negative	14	FHENumpy	115.115 ms	443.248 ms	924.341 ms
				Numpy	4.052 ms	3.003 ms	5.129 ms
6	ceil	Round to infinity towards positive	15	FHENumpy	116.452 ms	445.138 ms	946.710 ms
				Numpy	3.851 ms	2.967 ms	5.852 ms
7	trunc	Returns the truncated value of the input element	14	FHENumpy	114.795 ms	435.043 ms	920.229 ms
				Numpy	3.950 ms	3.051 ms	8.379 ms

Addition, multiplication, and subtraction functions on an axis

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to perform 10,000 operations)	
					Vector (length 10)	Vector (length 100)
1	prod	Returns the product of array elements over a given axis	16	FHENumpy	94.281 ms	189.424 ms
				Numpy	13.067 ms	13.588 ms
2	sum	Sum of array elements along a given axis	15	FHENumpy	93.359 ms	170.779 ms
				Numpy	14.384 ms	13.476 ms
3	nanprod	Returns the product of the array elements along a given axis, treating not-a-number (NaN) as 1	15	FHENumpy	94.679 ms	191.859 ms
				Numpy	31.317 ms	31.275 ms
4	nansum	Returns the sum of array elements along a given axis, treating not-a-number (NaN) as 0	15	FHENumpy	92.251 ms	171.722 ms
				Numpy	31.303 ms	31.124 ms
5	cumprod	Returns the cumulative product of the elements along the given axis	15	FHENumpy	110.025 ms	452.587 ms
				Numpy	7.684 ms	9.416 ms
6	cumsum	Returns the cumulative sum of elements along a given axis	15	FHENumpy	103.589 ms	184.747 ms
				Numpy	7.953 ms	9.212 ms
7	nancumprod	Returns the cumulative product of array elements along a given axis, treating not-a-number (NaN) as 1.	15	FHENumpy	108.990 ms	446.855 ms
				Numpy	24.041 ms	26.750 ms
8	nancumsum	Returns the cumulative sum of array elements along a given axis, treating not-a-number (NaN) as 0	15	FHENumpy	98.870 ms	183.906 ms
				Numpy	23.364 ms	25.846 ms
9	diff	Computes the nth discrete difference along a given axis	15	FHENumpy	100.214 ms	183.793 ms
				Numpy	14.953 ms	16.766 ms
10	ediff1d	Difference between consecutive elements of an array	15	FHENumpy	101.143 ms	183.307 ms
				Numpy	7.834 ms	8.713 ms
11	gradient	Returns the gradient of an N-dimensional array	15	FHENumpy	109.056 ms	417.985 ms
				Numpy	62.985 ms	62.642 ms
12	trapz		15	FHENumpy	95.977 ms	177.682 ms

		Integrate along a given axis using the composite trapezoidal rule		Numpy	32.875 ms	32.973 ms

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to perform 10,000 operations)	
					Vector (length 2)	Vector (length 3)
13	cross	Returns the cross product of two vectors	14	FHENumpy	126.352 ms	113.939 ms
				Numpy	90.003 ms	133.906 ms

Exponentials and Logarithms

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to perform 10,000 operations)		
					Scalar	Vector (length 10)	Vector (length 100)
1	exp	Exponential function	14	FHENumpy	145.605 ms	213.470 ms	623.747 ms
				Numpy	5.783 ms	3.110 ms	6.028 ms
2	expm1	Calculate $e^x - 1$ for all elements	14	FHENumpy	145.143 ms	216.903 ms	624.682 ms
				Numpy	5.260 ms	3.231 ms	10.724 ms
3	exp2	Compute 2^x of all elements	15	FHENumpy	142.820 ms	216.291 ms	624.373 ms
				Numpy	5.239 ms	6.357 ms	38.404 ms
4	expit	Calculate the sigmoid function	16	FHENumpy	143.406 ms	215.993 ms	630.411 ms
				Numpy	--	--	--
5	log	Natural logarithm	14	FHENumpy	100.188 ms	120.716 ms	214.483 ms
				Numpy	6.265 ms	3.595 ms	6.499 ms
6	log10	Logarithmic function to base 10	14	FHENumpy	99.330 ms	121.131 ms	215.899 ms
				Numpy	5.672 ms	3.271 ms	6.086 ms
7	log2	Logarithmic function to base 2	14	FHENumpy	99.975 ms	122.549 ms	215.778 ms
				Numpy	5.737 ms	3.265 ms	8.146 ms

Comparison Function

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to perform 10,000 operations) (vector length 100)		
					Scalar, scalar	Vector, Vector	Scalar, vector
1	compare	Compares the size of two input elements	12	FHENumpy	62.528 ms	272.686 ms	354.825 ms
				Numpy	--	--	--

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to perform 10,000 operations) (vector length 100) Unit: ms							
					Scalar, Scalar	Vector, Vector	Matrix, matrix	Scalar, vector	Scalar, matrix	Scalar, plain text	Vector, plain text	Matrix, plain text
2	equal (==)	Compares whether two input elements are equal	12	FHENumpy	61.380	277.862	385.457	370.251	627.629	48.669	177.992	241.455
				Numpy	5.753	3.080	3.159	5.339	5.408	5.259	5.440	5.347
3	not_equal (!=)	Compares whether two input elements are equal	11	FHENumpy	60.413	277.418	399.555	376.210	627.627	48.778	179.461	246.067
				Numpy	5.787	3.582	3.033	5.897	5.914	5.309	5.652	5.382
4	greater_equal (>=)	Determines whether the first input element is not less than the second input element	10	FHENumpy	59.843	280.780	409.738	379.432	620.931	48.862	179.423	244.288
				Numpy	5.602	3.013	3.172	5.837	5.728	5.753	5.458	5.807
5	less_equal (<=)	Determines whether the first input element is not greater than the second input element	11	FHENumpy	60.723	275.791	401.906	379.922	626.595	48.986	177.933	252.804
				Numpy	5.564	3.599	3.463	5.776	5.993	5.409	5.436	5.464

6	greater (>)	Determines whether the first input element is greater than the second input element	10	FHENumpy	63.545	281.271	412.159	388.317	631.072	48.995	177.199	243.294
				Numpy	5.561	3.449	3.146	5.817	5.688	5.743	5.330	5.479
7	less (<)	Determines whether the first input element is less than the second input element	11	FHENumpy	60.649	276.143	382.917	370.814	616.863	48.200	177.310	243.201
				Numpy	5.354	3.091	3.089	5.548	5.652	5.377	5.370	5.891

Miscellaneous

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to perform 10,000 operations)		
					Scalar	Vector (length 10)	Vector (length 100)
1	sqrt	Returns the non-negative square root of the input element.	16	FHENumpy	90.064 ms	104.204 ms	191.599 ms
				Numpy	5.838 ms	3.088 ms	6.618 ms
2	cbrt	Returns the cube root of the input element	15	FHENumpy	575.615 ms	604.566 ms	685.186 ms
				Numpy	5.462 ms	6.390 ms	40.365 ms
3	square	Returns the square of the input element.	16	FHENumpy	80.484 ms	99.163 ms	181.118 ms
				Numpy	4.984 ms	4.335 ms	4.706 ms
4	absolute	Returns the absolute value of the input element	16	FHENumpy	81.066 ms	97.004 ms	177.916 ms
				Numpy	3.599 ms	2.419 ms	2.720 ms
5	sign	Returns the element-wise indication of the sign of a number	16	FHENumpy	82.657 ms	101.188 ms	194.800 ms
				Numpy	4.004 ms	2.487 ms	3.147 ms
6	interp	One-dimensional linear interpolation	14	FHENumpy	210.396 ms	247.732 ms	606.056 ms
				Numpy	8.353 ms	8.243 ms	11.552 ms

7	nan_to_num	Replace nan with zero and inf with large finite numbers	--	FHENumpy	80.291 ms	96.725 ms	179.486 ms
				Numpy	6.067 ms	56.982 ms	58.664 ms
8	sort	Sort the elements of a given array	--	FHENumpy	--	99.282 ms	220.001 ms
				Numpy	--	10.254 ms	14.121 ms
9	clip	Limiting the values in an array	16	FHENumpy	109.241 ms	214.259 ms	589.787 ms
				Numpy	18.916 ms	5.397 ms	6.609 ms

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to perform 10,000 operations) (vector length 100)		
					Scalar, scalar	Vector, Vector	Scalar, vector
10	heaviside	Calculate the Heaviside step function	15	FHENumpy	96.774 ms	300.862 ms	381.637 ms
				Numpy	7.932 ms	7.514 ms	9.655 ms
11	maximum	Retrieve the largest element from the array	16	FHENumpy	96.078 ms	302.517 ms	377.158 ms
				Numpy	6.607 ms	7.960 ms	10.531 ms
12	minimum	Take the smallest element in the array	16	FHENumpy	96.313 ms	305.075 ms	381.020 ms
				Numpy	5.547 ms	7.311 ms	9.563 ms
13	fmax	Take the largest element from two arrays	16	FHENumpy	96.965 ms	308.925 ms	382.682 ms
				Numpy	5.793 ms	4.800 ms	7.303 ms
14	fmin	Take the smallest element from two arrays	16	FHENumpy	95.314 ms	306.556 ms	386.937 ms
				Numpy	5.583 ms	4.756 ms	7.306 ms
15	isclose	Determine whether array elements are equal within the error range	--	FHENumpy	64.954 ms	288.237 ms	378.020 ms
				Numpy	107.722 ms	89.825 ms	106.728 ms

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average single time to perform 10,000 operations) (vector100 and vector100)		
					mode：full	mode：same	mode：valid
16	convolve	Returns the discrete linear convolution of two 1D sequences	15	FHENumpy	451.736 ms	406.950 ms	367.971 ms
				Numpy	42.796 ms	26.531 ms	12.197 ms

Linear Algebra

Matrix and vector products

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average single time to execute 10,000 operations) (vector length100, matrix size 10*10)			
					Scalar, scalar	Vector, Vector	Scalar, vector	Matrix, Matrix
1	dot	Computes the dot product of two arrays	16	FHENumpy	109.532 ms	279.578 ms	373.610 ms	--
				Numpy	6.209 ms	3.152 ms	5.639 ms	--
2	inner	Calculates the inner product of two arrays	16	FHENumpy	107.867 ms	274.794 ms	371.033 ms	--
				Numpy	6.544 ms	3.734 ms	6.493 ms	--
3	outer	Computes the outer product of two vectors	15	FHENumpy	--	1121.219 ms	--	--
				Numpy	--	80.692 ms	--	--
4	matmul (@)	Computes the matrix product of two arrays	15	FHENumpy	--	293.728 ms	--	511.812 ms
				Numpy	--	3.131 ms	--	5.387 ms
5	kron	Compute the Kronecke product of two matrices	16	FHENumpy	--	--	--	4291.798 ms
				Numpy	--	--	--	230.669 ms

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to execute 10,000 operations)
					Matrix (10*10)
6	(linalg.)matrix_power	Raises a matrix to power n (integer)	15	FHENumpy	1761.459 ms
				Numpy	23.786 ms

Norms and other numbers

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to execute 10,000 operations)		
					Vector (length 10)	Vector (length 100)	Matrix (10*10)
1	(linalg.)norm	Calculate the vector norm	14	FHENumpy	441.640 ms	538.264 ms	--
				Numpy	17.418 ms	16.655 ms	--
2	det	Calculate the determinant of an array	15	FHENumpy	--	--	334.740 ms
				Numpy	--	--	27.366 ms
3	trace	Compute the trace of a matrix	16	FHENumpy	--	--	232.596 ms
				Numpy	--	--	14.879 ms

Solving equations and inverting matrices

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to execute 10,000 operations)
					Matrix (10*10)
1	(linalg.)inv（.I）	Find the inverse of a matrix	16	FHENumpy	778.624 ms
				Numpy	50.645 ms

Miscellaneous

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to perform 10,000 operations)	
					Vector (length 10)	Vector (length 100)
1	vander	Construct a Vandermonde matrix, where each column of the matrix is a power of the input vector	15	FHENumpy	190.055 ms	1243.777 ms
				Numpy	29.029 ms	48.087 ms

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to perform 10,000 operations)	
					Vector, Vector (length 10)	Vector, Vector (length 100)
2	polyfit	Using the least squares method for polynomial fitting	11	FHENumpy	544.589 ms	2543.988 ms
				Numpy	229.598 ms	469.536 ms

Statistics

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average time to perform 10,000 operations)	
					Vector (length 10)	Vector (length 100)
1	min	Returns the minimum value of the array	16	FHENumpy	93.150 ms	173.834 ms
				Numpy	17.057 ms	15.071 ms
2	max	Returns the maximum value of the array	16	FHENumpy	92.404 ms	172.180 ms
				Numpy	14.006 ms	14.245 ms
3	nanmin	Returns the minimum value of the array, ignoring any NaNs	16	FHENumpy	93.991 ms	176.304 ms
				Numpy	28.076 ms	27.959 ms
4	nanmax	Returns the maximum value of the array, ignoring any NaNs	16	FHENumpy	94.339 ms	176.385 ms
				Numpy	27.872 ms	28.540 ms
5	ptp	the range of values along the axis	15	FHENumpy	92.994 ms	173.347 ms
				Numpy	30.726 ms	30.749 ms
6	percentile	Computes the qth percentile of the data along the specified axis	16	FHENumpy	99.357 ms	217.666 ms
				Numpy	296.393 ms	300.741 ms
7	nanpercentile	Computes the qth percentile of the data along the specified axis while ignoring NaN values.	16	FHENumpy	98.963 ms	220.307 ms
				Numpy	305.241 ms	310.523 ms
8	quantile	Computes the qth quantile of the data along the specified axis	16	FHENumpy	99.163 ms	215.321 ms
				Numpy	295.489 ms	313.244 ms

9	nanquantile	Computes the qth quantile of the data along the specified axis while ignoring NaN values.	16	FHENumpy	98.361 ms	219.446 ms
				Numpy	311.907 ms	311.302 ms
10	median	Compute the median along the specified axis	16	FHENumpy	93.313 ms	211.660 ms
				Numpy	114.272 ms	117.412 ms
11	nanmedian	Compute the median along the specified axis, ignoring any NaNs	16	FHENumpy	93.785 ms	216.449 ms
				Numpy	138.605 ms	141.463 ms
12	average	Computes the weighted average along the specified axis	16	FHENumpy	94.451 ms	174.631 ms
				Numpy	27.092 ms	24.723 ms
13	mean	Computes the arithmetic mean along the specified axis	16	FHENumpy	92.742 ms	173.869 ms
				Numpy	24.386 ms	24.150 ms
14	nanmean	Compute the arithmetic mean along the specified axis, ignoring any NaNs	16	FHENumpy	92.176 ms	176.403 ms
				Numpy	95.066 ms	97.999 ms
15	std	Computes the standard deviation along the specified axis	16	FHENumpy	102.480 ms	188.203 ms
				Numpy	77.936 ms	78.011 ms
16	nanstd	Compute the standard deviation along the specified axis, ignoring any NaNs	16	FHENumpy	105.160 ms	193.704 ms
				Numpy	214.556 ms	216.497 ms
17	var	Computes the variance along the specified axis	15	FHENumpy	93.393 ms	182.906 ms
				Numpy	65.665 ms	65.847 ms
18	nanvar	Computes the variance along the specified axis, ignoring any NaNs	15	FHENumpy	95.924 ms	186.082 ms
				Numpy	213.534 ms	201.469 ms

No.	**Function Name**	**Function Description**	**Number of significant digits**	**Contrast to Numpy**	**Speed (average time to execute 10,000 operations)**
					Matrix (10*10)
19	corrcoef	Returns the Pearson product-moment correlation coefficient	16	FHENumpy	14996.567 ms
				Numpy	206.306 ms
20	cov	Returns the estimated covariance matrix	16	FHENumpy	3142.248 ms
				Numpy	142.949 ms

No.	Function Name	Function Description	Number of significant digits	Contrast to Numpy	Speed (average single time to perform 10,000 operations) (vector100 and vector100)		
					mode：full	mode：same	mode：valid
21	correlate	Returns the cross correlation of two 1-D sequences	16	FHENumpy	703.403 ms	698.743 ms	695.160 ms
				Numpy	5.186 ms	20.278 ms	5.106 ms

No.	Function Name	Function Description	Accuracy	Contrast to Numpy	Speed (average time to perform 10,000 operations)	
					Vector, Vector (length 10)	Vector, Vector (length 100)
22	digitize	Returns the index of the bin to which each value in the input array belongs.	100%	FHENumpy	105.401 ms	334.083 ms
				Numpy	3.408 ms	10.541 ms

Utility Functions

No.	Function Name	Function Description	Contrast to Numpy	Speed (average single execution time for 10,000 operations) (vector length 100, matrix size 10*10)
1	empty	Create an empty ciphertext scalar	FHENumpy	67.493 ms
			Numpy	3.367 ms
2	empty_array	Create an empty ciphertext vector	FHENumpy	72.879 ms
			Numpy	3.312 ms
3	ones	Return 1 scalar ciphertext	FHENumpy	62.575 ms
			Numpy	67.957 ms
4	ones_like	Create a ciphertext array of all 1s with the same shape as the input array	FHENumpy	89.894 ms
			Numpy	7.322 ms
5	zeros	Returns 0 scalar ciphertext	FHENumpy	61.668 ms
			Numpy	3.177 ms
6	zeros_like		FHENumpy	90.458 ms

		Create an all-0 ciphertext array with the same shape as the input array	Numpy	11.365 ms
7	eye	Create a unit ciphertext array of a given shape	FHENumpy	87.741 ms
			Numpy	8.931 ms
8	argmin	Returns the index corresponding to the minimum value	FHENumpy	134.582 ms
			Numpy	6.162 ms
9	argmax	Returns the index corresponding to the maximum value	FHENumpy	135.566 ms
			Numpy	6.026 ms
10	arange	generates a series of consecutive integers	FHENumpy	185.562 ms
			Numpy	3.386 ms
11	random.choice	Sampling function	FHENumpy	108.182 ms
			Numpy	10.745 ms

No.	**Function Name**	**Function Description**	**Contrast to Numpy**	**Speed (average time to execute 10,000 operations)**	
				Vector (length 100)	**Matrix (size 10*10)**
12	ones_array	Creates an all-1 ciphertext array of a given shape	FHENumpy	75.393 ms	83.468 ms
			Numpy	7.757 ms	6.043 ms
13	zeros_array	Create an all-zero ciphertext array of a given shape	FHENumpy	73.747 ms	83.180 ms
			Numpy	3.403 ms	1.585 ms
14	argsort	Sort the array and extract the index output corresponding to it before sorting	FHENumpy	209.259 ms	227.797 ms
			Numpy	11.998 ms	10.349 ms
15	random.rand	Randomly generate an array of a given shape, where each element value is between 0 and 1.	FHENumpy	200.065 ms	218.655 ms
			Numpy	9.195 ms	9.328 ms
16	flip	Reverses the order of array elements along the specified axis	FHENumpy	188.735 ms	249.121 ms
			Numpy	3.821 ms	4.068 ms
17	where	Select elements from multiple selections based on conditions	FHENumpy	185.497 ms	256.140 ms
			Numpy	8.304 ms	11.784 ms

No.	Function Name	Function Description	Contrast to Numpy	Speed (average time to execute 10,000 operations)	
				Scalar, vector (vector length 100)	Vector, Vector (vector length 100)
18	append	Concatenate arrays and add some values to the original array	FHENumpy	204.003 ms	315.390 ms
			Numpy	16.203 ms	14.534 ms

Array Operations

No.	Function Name	Function Description	Contrast to Numpy	Speed (average time to execute 10,000 operations)	
				Vector (length 100)	Matrix (size 10*10)
1	transpose（.T）	Find the transpose of an array	FHENumpy	--	530.181 ms
			Numpy	--	2.603 ms
2	insert	Insert an element at a specified position in an array	FHENumpy	--	84.884 ms
			Numpy	--	57.318 ms
3	cipherLen	Returns the length of the input array	FHENumpy	4.364 ms	2.529 ms
			Numpy	0.500 ms	0.422 ms
4	cipherShape	Returns the shape of the input array	FHENumpy	4.683 ms	3.425 ms
			Numpy	1.100 ms	1.105 ms